Evaluating the Multilingual Transferability of LLM-as-a-Judge Prompting Protocols

Submitted on: **23-02-2026**

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1 Introduction

1.1 Scientific Context and Problem

The introduction of Large Language Models (LLMs) has revolutionized the field of NLP fundamentally [5]. They are no longer only used as tools to generate text; instead, their use as evaluators of system outputs is becoming increasingly popular. The use of LLM-as evaluators, known as LLM-as-a-Judge, has gained popularity as an alternative to other metrics such as BLEU and ROUGE, especially when it comes to open-ended generation output [19]. However, this brings with it the problem of ensuring the reliability, stability, and fairness of such evaluations, especially when carried out in different languages.

One of the main challenges associated with the use of LLMs as a-Judge for evaluation is the fact that LLMs are sensitive to prompt design [15], where even minor variations in prompts can result in significantly different evaluation results. Although strategies like self-consistency and Chain-of-Thought improve reasoning, particularly when used in English [17, 18], their effectiveness and performance in other languages remain unknown [10].

This problem is essentially important in relation to languages with fewer resources and greater linguistic diversity. For example, the performance of multilingual LLMs is known to degrade significantly when it comes to morphologically richer or less-represented languages[4]. Languages such as many African languages, Catalan, and Persian differ substantially in script and structure from high-resource languages such as English, which could interact with prompt design [6]. Despite the fact that initiatives such as Masakhane and datasets such as FLORES-200 have expanded multilingual coverage [13, 16], systematic analysis of prompt transferability across languages remains limited.

1.2 Research Gap

Large Language Models have been used as automatic evaluation tools in NLP tasks with the introduction of LLM-as-a-Judge models, which provide scalable solutions to traditional evaluation systems.[19]. Although recent studies have identified reliability issues with LLM-as-a-Judge models, including position bias, response length bias, and unstable judge behavior [10, 15], they have been conducted on English and other high-resourced languages.

In parallel, Prompt engineering plays a significant role when it comes to LLM's performance, demonstrating the effectiveness of strategies like Chain-of-Thought and self-consistency in improving the effectiveness of evaluation quality in English [17, 18, 20]. Nonetheless, it is still unclear how effective these advances can be across other languages with different scripts, grammatical and morphological structures, and resource availability. Therefore, even though multilingual benchmarks have expanded the scope of language evaluation systems [2, 3], there has been no study on the effectiveness of prompt engineering on LLM-as-a-Judge models across languages.

In particular, when we shift from high-resource languages like English to other lower-resource languages, such as many African languages or other morphologically complex languages, we find linguistic variation that interacts with prompt sensitivity in ways that are difficult to predict. To date, there is no unified framework that examines the performance of prompting strategies across languages, nor which linguistic factors influence judge reliability most.

We therefore do not know whether different prompting strategies that enhance LLM-as-a-Judge performance for English can be reliable when they are transferred to multilingual evaluation scenarios. Hence, the main objective of this thesis is to address these gaps by examining different prompting strategies across multiple languages within a state-of-the-art LLM-as-a-Judge framework. The primary goal is to determine whether approaches that are effective in high-resource languages can be reliably transferred to other linguistic contexts.

1.3 Proposed Contribution

The main goal of this thesis is to provide a systematic benchmark of different prompting strategies for LLM-as-a-Judge evaluation in a multilingual context. Specifically, this research proposes three main contributions. First, we evaluate the effectiveness of the prompting strategies to identify the most effective prompting methods that produce the most reliable LLM-as-a-Judge outputs. Second, we compare the effectiveness of the prompting strategies on both high-resource languages, such as English and low-resource languages (African and other low-resource languages) to identify if the effectiveness of the prompting strategies changes based on the language resource availability. Third, we identify the systematic failure patterns of the prompting strategies through a multilingual error analysis to identify the factors that reduce LLM-as-a-Judge effectiveness.

1.4 Research Questions

Main RQ: *To what extent do different prompting strategies impact the reliability of LLM-as-a-Judge systems in multilingual settings, and how does their effectiveness vary across languages with different resource levels and linguistic characteristics?*

Expanding the concept, we can lead into the following research **Sub-questions:**

- **RQ1:** How trustworthy are various prompting strategies when used in LLMs as judges, considering both high-resource and low-resource languages? Are some techniques more robust in handling linguistic variations than others?
- **RQ2:** Which linguistic properties are more related to variations in LLM-as-a-Judge reliability, and how does the effectiveness of a prompting strategy differ among languages with distinct scripts and grammatical structures?
- **RQ3:** What kind of failures do each prompting technique produce in different languages and are the error types language-dependent or universal?

2 Related Work

This thesis gathers three related areas of research: LLM-as-a-Judge systems for evaluation, methods, prompt engineering strategies, and the challenges of multilingual NLP. We review these topics, identify key gaps, and position our work with respect to the current state of the art.

2.1 LLM-as-a-Judge Evaluation

This line of work involves using LLMs as evaluators of other models. A comprehensive study of LLM judges using the frameworks of MT-Bench and Chatbot Arena by Zheng et al. [19] demonstrated the viability of using LLM judges for the evaluation of machine translation and summarization. Following this study, Li et al. [10] summarized the current understanding of the factors that help or challenge LLM-as-a-Judge evaluations. For instance, the authors emphasised the factors that influence the reliability of the LLM-as-a-Judge: the complexity of the task, the quality of the reference data, and the clarity of the prompt. However, the reliability of LLM-as-a-Judge is highly dependent on the prompt and instruction formulation. Shi et al. [15] demonstrated that LLM judges are subject to position bias, which occurs when the order of items in a comparative evaluation has a significant impact on the outcomes. This finding proposes that LLM judges are unstable under different prompting strategies, which raises essential questions about the generalizability and reproducibility of evaluation outputs done by LLM judges. Hence, recent work demonstrates that specific prompting strategies could vary the reliability of LLM Judges, yet systematic comparison of these strategies remains limited, especially in multilingual settings.

2.2 Prompt Engineering and Chain-of-Thought Strategies

Recent advances in prompt engineering have demonstrated how such instructions can potentially influence LLMs and their reasoning mechanisms. For example, Brown et al.'s [5] few-shot prompting method demonstrated how simply inputting examples can improve LLM performance significantly. Following this, Wei et al.'s [18] Chain-of-Thought prompting method demonstrated how prompting LLMs to outline a series of steps before arriving at an answer can significantly improve LLM performance.

Wang et al.'s[17] self-consistency method extended Chain-of-Thought prompting by having LLMs create multiple independent reasoning paths and then combining them to improve LLM performance. Zhou et al.'s [20] least-to-most prompting method extended Chain-of-Thought prompting by having LLMs solve problems by breaking them into simpler sub-problems and then solving them sequentially. Liu et al.'s [12] generated knowledge prompting method extends Chain-of-Thought prompting by having LLMs create knowledge before arriving at an answer. All these strategies have been demonstrated to be effective in improving LLM performance in high-resource languages such as English, suggesting that they may also enhance the reliability of LLM-based evaluation.

However, what is not clear is how such strategies would improve judge reliability when applied to a broader, more diverse range of languages. For example, would such strategies improve judge reliability when applied to languages, particularly those with different morphological properties, script systems, or lower training data representation? A key research gap is to discover how prompting strategies that enhance judge reliability in high-resource languages perform when applied to multilingual contexts.

2.3 Multilingual NLP and Low-Resource Language Challenges

Multilingual NLP models have seen significant growth with the introduction of models such as mBERT, mT5, and Llama 2. The Masakhane initiative [13], for instance, has been instrumental in developing machine translation resources for African languages. Similarly, the FLORES-200 corpus [16] provides high-quality translation pairs for over 200 languages, enabling systematic evaluation of multilingual systems. In parallel, Gehrmann et al. [9] introduced SEAHORSE, a multilingual summarization evaluation benchmark that includes human quality judgments across different languages.

Nevertheless, most of these benchmarks still emphasize on languages with high-resources such as English, and the gap in systematically evaluating LLM-as-a-Judge systems across a broader multilingual spectrum remains unclear. Recent studies have shown that multilingual LLMs often perform significantly worse on morphologically complex and low-resource languages [4]. Furthermore, Chung et al. [6] investigated the cross-lingual stability of LLM judges and found that reliability can vary substantially depending on script families and morphological characteristics. Despite progress and research in these areas, no study has systematically connected prompting strategies with multilingual judge reliability, specially in a low resource scenario, under controlled experimental conditions.

2.4 Cross-Lingual Transfer and Research Gap

Recent studies on cross-lingual transfer have revealed that knowledge can be transferred from high-resource languages to lower-resource ones, but this transfer is not achieved naturally and depends on linguistic similarity and task characteristics [14]. The question of cross-lingual prompt steerability—whether prompts can consistently guide LLM behavior across different languages—remains largely unexplored [1]. While holistic evaluation frameworks have begun addressing the multi-dimensional nature of LLM assessment [11], multilingual fairness metrics for LLM-based evaluators are still underdeveloped.

The gap in the literature is clear: although prompt engineering has been shown to improve reasoning in English, and multilingual NLP challenges are well documented, there is limited knowledge and systematic evidence on how different prompting strategies affect LLM judge reliability across languages as well as which linguistic factors predict performance degradation.

Several strands of research address parts of this problem but leave key questions unanswered. LLM-as-a-Judge studies demonstrate that prompting significantly affects evaluation outcomes, yet they focus primarily on English settings [10, 15, 19]. Prompt engineering research shows that strategies such as Chain-of-Thought and self-consistency also improves reasoning performance in English [12, 17, 18, 20]. Yet, their cross-lingual effectiveness is unknown.. Multilingual NLP research highlights performance degradation in low-resource and morphologically complex languages [4, 6], yet does not systematically assess how prompting strategies behave in these contexts. Similarly, cross-lingual transfer research investigates model transferability [1, 14], but does not examine LLMs in their role as evaluators.

This thesis bridges these lines of research by systematically comparing five prompting strategies—zero-shot/few-shot prompting, Chain-of-Thought prompting, self-consistency prompting, least-to-most prompting, and generated knowledge prompting—across multiple languages spanning diverse resource levels and grammatical properties. Our analysis builds upon established evaluation benchmarks, including judge–human agreement baselines in English [19], multilingual evaluation frameworks with human judgments [9], cross-lingual judge stability studies [6], and correlation-based evaluation protocols [8]. Using these studies as reference points, this research aims to provide a systematic cross-lingual evaluation of prompting strategy effectiveness for LLM-as-a-Judge systems.

3 Methodology

This thesis design employs a controlled experimental setup where the prompt design is the primary variable of interest. The research aims to identify how different prompt engineering strategies impact the reliability, stability, and cross-lingual transferability of using LLMs as judges for evaluating model-generated output across different tasks. This design follows recent research that identified that LLMs are highly prompt-sensitive and that this sensitivity has important implications for downstream evaluation tasks [10, 15, 19].

3.1 Ground Truth and Evaluation Data

The evaluation is based on existing multilingual benchmarks that have associated human judgments on how well the model's output performs. Depending on the specific task and dataset used, these human judgments may be presented as scalar quality scores, rankings, or pairwise preferences [8, 9]. For African languages, the study relies on resources developed by the Masakhane initiative, such as MasakhaNER, MasakhaNEWS, and MasakhaPOS [13]. and AfriQA, which provides gold labels and reference annotations. For other multilingual languages like Catalan and Persian, we are using other popular benchmarks like FLORES-200 for machine translation tasks [16] and multilingual NLI benchmarks like XNLI, where gold labels are available [7]. These annotations serve as the ground truth reference for evaluating LLM-as-a-Judge outputs and are normalized to a common format to make cross-lingual comparison possible.

3.2 Experimental Conditions

3.2.1 Fixed Conditions. To evaluate the effect of different prompting strategies, the following non-prompt variables are held constant across different experimental conditions:

- Evaluated system outputs, which are pre-generated and frozen before evaluation
- LLM inference parameters, including decoding strategy and temperature
- Human reference judgments used for comparison

Providing a controlled setup ensures that any change in evaluation outcomes is due solely to differences in prompting strategy, depending on each task and language. For all experiments, open-source large language models are used as judges without any task-specific fine-tuning. This means that these models are treated as an evaluation tool rather than objects of comparison [19]. Additionally, in all experiment, it is ensured that all inference parameters are remained same.

3.2.2 Prompting Strategies. As mentioned previously, the primary experimental variable is the prompting strategy used to elicit judgments from LLMs. Drawing on recent advances in prompt engineering, we evaluate strategies including: zero-shot and few-shot prompting [5], Chain-of-Thought prompting [18], self-consistency approaches [17], least-to-most prompting [20], and generated knowledge prompting [12].

Critically, we study these strategies in a multilingual setting across different tasks and languages [1, 14]. This design enables systematic analysis of whether prompting techniques that improve reasoning or stability in high-resource English settings transfer reliably to other languages, scripts, and morphologically complex languages [4, 6]. Each prompting strategy is applied under identical experimental conditions to isolate the effects of prompt design.

3.3 Judge Evaluation and Reliability Assessment

We measure how reliable LLMs are as automatic judges by using established meta-evaluation approaches [8, 10, 19]. Several open-source LLMs is tested, both in single- and cross-lingual settings, using the constructed multilingual benchmark.

In our experiments, human-annotated reference texts and labels from selected datasets act as ground truths for evaluation. For generative tasks, the LLM judges compare their system-generated output with corresponding reference texts. We then verify how well these rankings align with the expected quality order between better- and worse-performing models.

Our primary evaluation measure is Spearman Rank Correlation (ρ_s), as it is standard for ordinal evaluation signal assessment, aligning with earlier studies on LLM as a Judge evaluation.[8, 19].

3.4 Validation and Comparative Analysis

We measure the judge reliability through a combination of quantitative measures and qualitative analysis, using established LLM-as-a-Judge evaluation practices[15, 19]. First, we conduct an error

analysis with the idea of identifying recurring errors and failures, including cultural misinterpretations, script-related issues, code-switching, and differences in dialect. This will allow us to identify whether certain prompting methodologies contain bias or degrade under specific linguistic scenarios [6].

For the quantitative evaluation, we compare LLM judge outputs with human ground truth annotations. We then validate by using Spearman's rank correlation (ρ_s), which is the most common evaluation approach when correlation rankings are used [8]. The correlation coefficients are computed to assess the reliability and consistency across languages and methods. Where appropriate, the confidence intervals are reported, ensuring robust comparison within different conditions. We then evaluate results across different languages, resource levels, and tasks using agreement scores to see which approaches remain effective and where performance declines.

4 Risk Assessment

Although the proposed methodology is realistic within the given time frame, certain risks could affect the pace of the project. Hence, in this section, the main challenges and their corresponding mitigation strategies are addressed.

The first major concern is the quality of the data and the coverage of the languages. It has been noticed that multilingual evaluation sets are not always balanced, especially in the case of low-resource languages. It is possible that the quality of the annotation could be inconsistent across different evaluation sets. To address this concern, the study plans to be flexible with the data selection strategy supported by a modular evaluation pipeline. This allows language-task combinations to be easily adjusted if the quality of data proves insufficient. Also, where needed, the language set could be reduced while preserving diversity in script types and resource levels. The idea is to start with African languages and if the data is available, expand it to other languages such as Persian or Catalan.

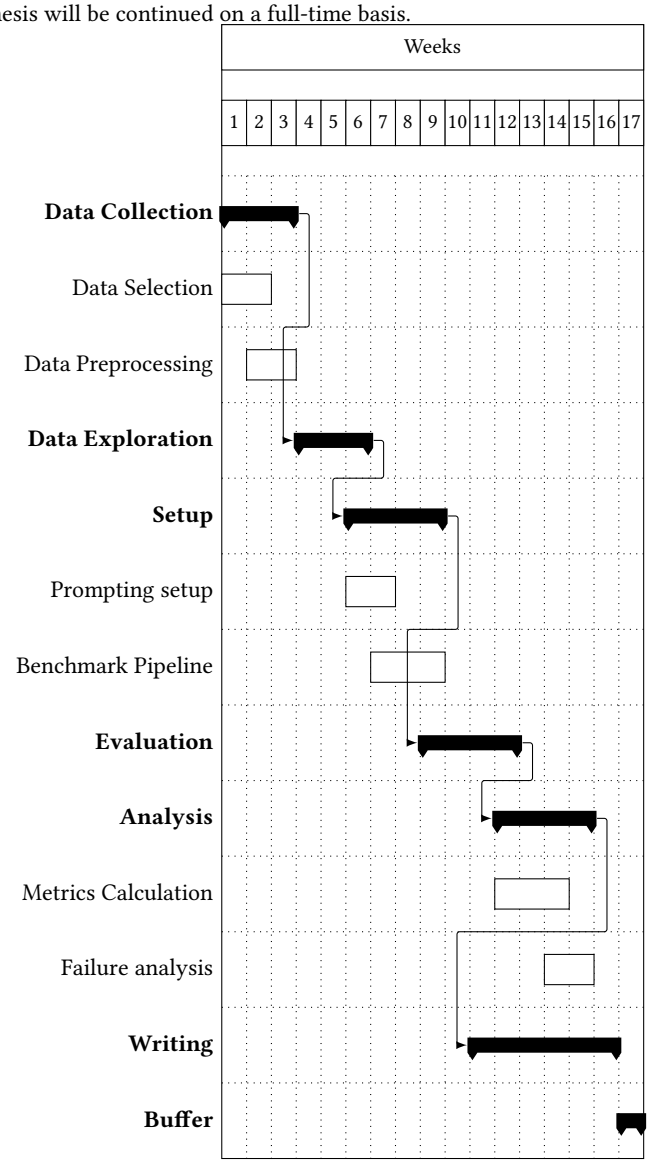
The second major concern is the computational requirement. It is possible that testing different prompting methods on different languages and conducting repeated experiments to determine the stability could become computationally challenging and demanding. To address this concern, it is planned to ensure that the experiments are within reasonable limits. Hence, to start, the study aims to test a limited set of models to keep the experiments computationally manageable.

A third challenge arises from the nature of the instability of the LLM-as-a-Judge approach. The prior study indicated that the evaluation results may change over time and are affected by the prompt. This may affect the interpretation of the results in the case of cross-lingual comparisons. We will mitigate this by running the evaluation multiple times and using variance-aware metrics, the coefficient of variation, alongside agreement measures. This approach can reduce the effect of stochastic noise.

Lastly, the scope is an essential factor to consider. The combinatorial nature of the study, which includes the use of multiple languages and prompting strategies, may result in a risk of extension. We will mitigate this by using a step-by-step approach. We will start with a specific and well-defined pipeline and then expand to include additional aspects if time and the results allow. This will ensure that the main research questions are adequately addressed within the thesis time frame.

5 Project Plan

The project plan is shown in the Gantt chart below. The thesis has seven tasks which are divided into six main phases, taking in total 17 weeks to complete. The planning starts with data collection and exploration, followed by the setup of the experiment and finally, the evaluation, benchmarking, and analysis. Alongside the phase involves data collection and exploration, followed by the setup of the experiment, and finally the evaluation and analysis of the data. Alongside the final phases, writing will be done and the last week is kept as a buffer to ensure the thesis is done on time. The first six weeks of the project are scheduled to be part-time, after which the thesis will be continued on a full-time basis.



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